

Supplementary Information

1. Study dataset and analytical coverage

Sample sizes varied across analyses according to data availability and analytical scale. The national study frame comprised 18 cities, while the geocoded garden-level inventory included 291 CGs across 16 cities; no CGs were reported in Bnei Brak, and we could not obtain garden-level data from Ashkelon (see Supplementary Dataset S1 and Dataset S2 on GitHub at <https://github.com/galiacu/Making-Community-Gardens-Count>). We used smaller complete-case samples where required. Missing questionnaire-derived garden-level data were not imputed; the two spatial SEI assignments are described in the Methods and Supplementary Notes. Supplementary Table S1 summarises the sample used in each analysis.

Supplementary Table S1 | Analytical samples and data availability across study components

Analysis	Gardens (n)	Cities (n)
Full geocoded inventory	291	16
Municipal governance questionnaire	NA	17
Formally designated green-space–satisfaction analysis	NA	18
CG location relative to formally designated green areas	291	16
Neighbourhood SEI distribution	291	16
Neighbourhood-to-city Δ SEI analysis	291	16
Annual-visit data available	214	13
Active-member-annual-visit analysis	200	13
Garden-level characteristics adjusted model	199	13
Funding sources adjusted model	214	13
Socioeconomic context adjusted model	214	13
Spatial/GIS context adjusted model	214	13
Community profile adjusted model	214	13
Common sample for AIC/BIC comparison	199	13

Note: NA indicates analyses conducted at the city level for which a garden-level sample size is not applicable. Dimension-specific ordinal-logit models used complete cases for the outcome and variables included in each model. The common sample used for AIC and BIC comparison comprised gardens with complete data across all variables required for the five dimension-specific models.

2. Analytical sample and completeness of annual-visit data

Supplementary Table S2 | Availability of annual-visit data by municipality. *The table reports the number of CGs in the study inventory with and without annual visit-category data in each municipality.*

City	Total CGs (n)	CGs with annual-visit data (n)	Missing annual-visit data (n)
Tel Aviv-Yafo	69	50	19
Rehovot	16	0	16
Netanya	18	6	12
Ramat Gan	10	0	10
Rishon LeZion	6	0	6
Ashdod	13	8	5
Kfar Sava	18	13	5
Haifa	15	12	3
Petah Tikva	4	3	1
Herzliya	14	14	0
Holon	9	9	0
Beer Sheva	22	22	0
Bat Yam	1	1	0
Beit Shemesh	1	1	0
Hadera	4	4	0
Jerusalem	71	71	0
TOTAL	291	214	77

Note: "CGs with annual-visit data" refers to gardens with a recorded level of the annual-visit ordinal variable. Missing values were not imputed. Annual-visit data were available for 214 of 291 gardens (73.5%); 77 gardens (26.5%) had no recorded annual-visit data.

3. Sensitivity analysis of aggregate engagement estimates

Active-member and annual-visit data were reported in ordinal categories. Annual-visit categories reflected respondent estimates because standardised direct visitor counts were unavailable. We then derived aggregate active-member and annual-visit estimates by assigning representative values to the reported categories. We used interval midpoints where applicable, and sensitivity analyses varied the representative values assigned to the lowest and highest categories. The resulting primary estimates were 3,180 active members across 227 gardens and 301,850 annual visits across 214 gardens with available data.

Supplementary Table S3 | Sensitivity analysis of aggregate active-member and annual-visit estimates under alternative representative values.

A. Active members

Scenario	Up to 10	10-20	20-30	30-40	Above 40	Estimated active members
Lower sensitivity	3	15	25	35	45	2,901
Primary	5	15	25	35	50	3,180
Higher sensitivity	8	15	25	35	60	3,631

B. Annual visits

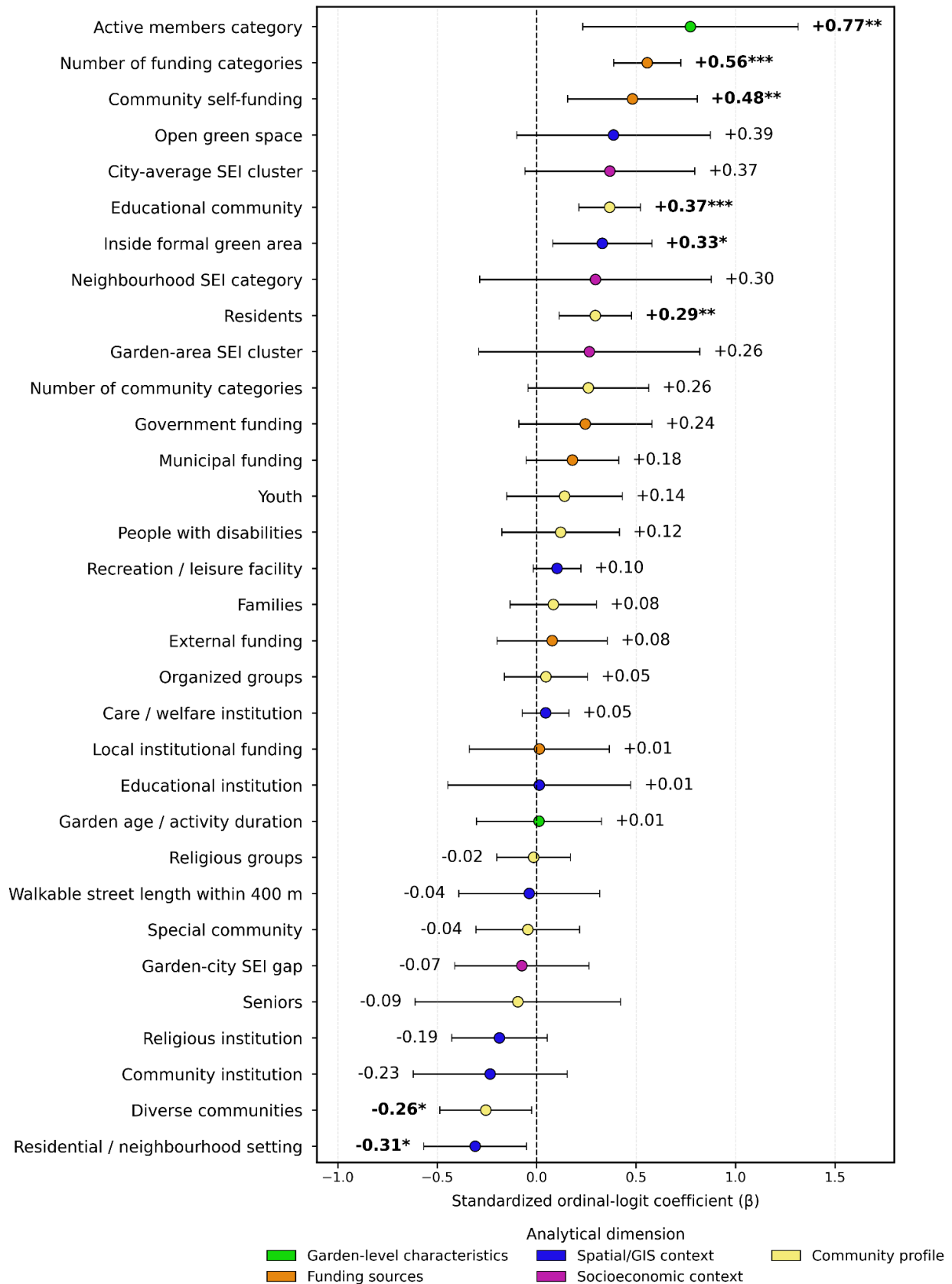
Scenario	Up to 100	100 - 500	500 - 1,000	1,000 - 5,000	5,000 - 10,000	Above 10,000	Estimated annual visits
Lower sensitivity	25	300	750	3,000	7,500	10,000	282,100
Primary	50	300	750	3,000	7,500	12,000	301,850
Higher sensitivity	75	300	750	3,000	7,500	15,000	330,600

Note: Primary estimates used representative values of 5 and 50 for the lowest and highest levels of the active-member ordinal variable, respectively, and 50 and 12,000 for the lowest and highest levels of the annual-visit ordinal variable. We calculated estimates only for gardens with available data and did not extrapolate to gardens with missing data.

4. Univariate ordinal regression

We first examined each variable in a univariate proportional-odds ordinal logistic regression model. Higher levels of the annual-visit ordinal variable were positively associated with the active-member variable ($\beta = 0.77$, $P = 0.009$), number of funding sources ($\beta = 0.56$, $P < 0.001$), community self-funding ($\beta = 0.48$, $P = 0.007$), educational-community orientation ($\beta = 0.37$, $P < 0.001$), location within a formally designated green area ($\beta = 0.33$, $P = 0.014$) and resident orientation ($\beta = 0.29$, $P = 0.004$), and negatively associated with residential/neighbourhood setting ($\beta = -0.31$, $P = 0.023$) and diverse-community orientation ($\beta = -0.26$, $P = 0.033$).

Remaining variables were not statistically significant; we then used dimension-specific models to assess within-dimension adjusted associations.



Supplementary Figure S1 | Univariate associations with annual visits. Standardised coefficients and 95% confidence intervals from separate univariate proportional-odds ordinal logistic regression models. Each model included a single variable. Positive coefficients indicate higher odds, and negative coefficients indicate lower odds, of belonging to a higher level of the annual-visit ordinal variable. The active-member model included 200 gardens, the garden-age/activity-duration model included 213 gardens, and the remaining models included 214 gardens; all analyses included 13 cities. Because gardens were nested within municipalities, we estimated coefficient uncertainty using municipality-clustered robust standard errors, allowing observations from the same city to be correlated. Given the relatively small number of municipal clusters in the visitor analysis (13 municipalities), we used finite-sample correction and t-based inference rather than large-sample normal inference. Colours denote the five analytical dimensions. Error bars indicate 95% confidence intervals. *P < 0.05, **P < 0.01, ***P < 0.001.

5. Relative fit of dimension-specific models

Each dimension-specific ordinal-logit model included the variables assigned to that analytical dimension. For direct comparison of model fit, we refitted all five models using the same complete-case sample of 199 gardens across 13 cities. The garden-level characteristics model had the lowest AIC and BIC (AIC = 609.06; BIC = 632.12), with $\Delta\text{AIC} > 20$ for each other dimension-specific model. Its Akaike weight was 0.9999, indicating substantially greater relative support than the other four dimension-specific models.

Design-matrix diagnostics indicated full rank for all five dimension-specific models, with condition numbers ranging from 1.15 to 2.27 and maximum absolute pairwise variable correlations ranging from 0.14 to 0.45, which do not indicate problematic multicollinearity.

Supplementary Table S4 | Relative fit of dimension-specific proportional-odds ordinal logistic regression models

Parameter	Garden-level characteristics	Funding sources	Spatial/GIS context	Socioeconomic context	Community profile
Variables	2	5	9	2	9
AIC	609.063	629.078	630.458	637.685	640.053
BIC	632.116	662.011	676.564	660.738	686.159
Log-likelihood	-297.531	-304.539	-301.229	-311.842	-306.027
Δ AIC	0.000	20.015	21.395	28.622	30.990
Akaike weight	0.99993	4.51×10^{-5}	2.26×10^{-5}	6.09×10^{-7}	1.86×10^{-7}

Note: All models were refitted using the same complete-case sample of 199 CGs across 13 cities to permit direct comparison. Lower AIC and BIC values indicate better relative fit after accounting for model complexity. ΔAIC is the difference from the minimum AIC, and Akaike weights quantify relative AIC-based support within the five-model comparison set and sum to 1.

6. Municipal governance and GIS representation

The table compares municipality-reported CG layers with layers located and examined during an independent audit of official municipal GIS portals or spatial datasets. It summarises municipal responsibility for CGs, layer geometry and accessibility, representation of the study inventory, classification within municipal GIS portals, and the spatial unit represented by polygon records.

Supplementary Table S5 | Municipal governance and independent GIS audit of CGs across the study cities.

City	Municipality reported GIS representation	Municipal unit responsible for CGs	CGs in the study inventory (n)	CG layer identified in GIS audit	CGs found in GIS audit (n)	The GIS layer can be downloaded	Parent category in the municipal GIS portal	CGs not found in GIS audit (n)	GIS inventory gap (%)	CG area represented by the polygon
Ashdod	Yes, Points	Environment	13	Point layer verified	8	YES	Recreation and sports	5	38%	
Ashqelon	No municipal response	No municipal response	No municipal response	No municipal response	NA			No Data	No Data	
Bat Yam	No layer	Community & Welfare	1	No layer	NA			1	100%	
Beer Sheva	Yes, Polygon	Environment & Welfare	22	Reported layer not located	NA			22	100%	Layer could not be found during independent verification
Beit Shemesh	No layer	External (Non-municipal organisation)	1	No layer	NA			1	100%	
Bene Beraq	No layer	Environment & Gardening	No CGs in the city	No layer	NA			No CGs	No CGs	
Hadera	No layer	Environment	4	No layer	NA			4	100%	
Haifa	Yes, Polygon	Community	15	Reported layer not located	NA			15	100%	Layer could not be found during independent verification
Herzliya	Yes, Polygon	Environment	14	Polygon layer verified	13	NO	Environment	1	7%	Polygon represents the parcel/site; notes indicate when the CG occupies only part of the mapped area.
Holon	Yes, Points	Environment	9	Reported layer not located	NA			9	100%	
Jerusalem	Yes, Polygon	Community	71	Polygon layer verified	44	YES	Culture and leisure	27	38%	Area values manually entered from municipal CG coordinator reports.
Kfar Sava	Yes, Polygon	Sustainability & Innovation	18	Polygon layer verified	15	NO	Parks, landscape and sustainability	3	17%	The polygon represents either the parcel or the garden extent, depending on

										the record.
Netanya	Yes, Points	Operations & Neighbourhood Administration	18	Reported layer not located	NA			18	100%	
Petah Tiqwa	No layer	Environment	4	No layer	NA			4	100%	
Ramat Gan	Yes, Points	City Improvement	10	Point layer verified	10	NO	Parks and landscape	0	0%	
Rehovot	No layer	Community	16	No layer	NA			16	100%	
Rishon LeZiyyon	No layer	Environment	6	No layer	NA			6	100%	
Tel Aviv	Yes, Points	Environment	69	Point layer verified	67	YES	Tourism, leisure, culture and sports	2	3%	
TOTAL	10 cities reported yes for CGs layer		291	4 cities didn't have a CGs layer as reported	157	6 cities with a CGs layer; 3 can be downloaded		134	46%	Inconsistent area indication in 3/5 polygon layers that were found

Note: Municipal GIS records were checked in July 2026. A CG was counted as represented in municipal GIS when it could be matched to a record in a municipal CG layer located during the independent audit. “CGs not found in GIS” includes gardens for which no corresponding record could be identified in the municipal GIS systems examined, including municipalities where a reported layer could not be located during the audit. This does not exclude the possibility that additional records or internal or non-public layers existed but were inaccessible at the time of the review. The audit assessed representation of sites as CGs in municipal spatial systems; it did not assess whether vegetation at these locations was detectable in land-cover or remote-sensing data. The GIS inventory gap represents the proportion of the corresponding study inventory that was not found in the municipal GIS systems examined. “Complete GIS layer downloadable” indicates that the full layer could be downloaded as a dataset rather than only viewed through the municipal GIS portal. The spatial-unit field applies only to polygon layers. Because polygon records did not follow a common boundary/footprint definition, we did not interpret polygon area as CG area or compare it across municipalities. NA denotes not applicable; “No data” indicates that the relevant information was unavailable; and “No municipal response” indicates that the municipality did not provide questionnaire data.

Supplementary Table S6 | Distribution of binary funding and spatial variables used in the analyses.

Analytical dimension	Variable	Full inventory, n/N (%)	Annual-visit sample, n/N (%)
Spatial/GIS context	Inside formally designated green area (statutory layer)	62/291 (21.3%)	47/214 (22.0%)
Funding sources	Municipal funding	286/291 (98.3%)	209/214 (97.7%)
	Community self-funding	106/291 (36.4%)	84/214 (39.3%)
	Government funding	57/291 (19.6%)	28/214 (13.1%)
	Local institutional funding	36/291 (12.4%)	23/214 (10.7%)
	External funding	31/291 (10.7%)	22/214 (10.3%)
Spatial/GIS context	Open green space (reported site type)	131/291 (45.0%)	96/214 (44.9%)
	Community institution	78/291 (26.8%)	52/214 (24.3%)

	Educational institution	59/291 (20.3%)	40/214 (18.7%)
	Residential / neighbourhood setting	60/291 (20.6%)	54/214 (25.2%)
	Care/welfare institution	23/291 (7.9%)	19/214 (8.9%)
	Religious institution	19/291 (6.5%)	17/214 (7.9%)
	Recreation/leisure facility	24/291 (8.2%)	19/214 (8.9%)

Note: Values indicate the number and percentage of gardens for which each binary characteristic was present. Site-type variables were coded as non-mutually exclusive binary indicators because a garden could be associated with more than one site type. The annual-visit sample comprises gardens with available annual-visit data; Supplementary Table S1 reports model-specific complete-case sample sizes.

SUPPLEMENTARY NOTES

Data-collection instruments and response categories

Municipal questionnaire

An English translation of the data-collection items from the original Hebrew municipal questionnaire is provided below. Introductory, contact, and closing text is omitted.

Respondent and municipal information

Email address:

City:

What is the official job title of the person responsible for CG activities?

Under which municipal division or department are CGs managed?

How many CGs currently operate in the city?

Municipal GIS

Do CGs appear as a separate layer in the municipal GIS? Response options were: no separate layer; yes, represented as points; or yes, represented by their area as polygons.

If CGs appear as a separate GIS layer, under which category are they listed in the municipal GIS portal?

The form included an optional open-text field for additional comments or information relevant to the study.

Garden-level data form

The standardised garden-level form requested the following information for each CG:

Garden name; street address; characteristics of the garden location (for example, park, public open space, school or other institutional setting); active community associated with the garden; special characteristics of the community; garden age/activity duration; number of regular active members; estimated annual visits; neighbourhood socioeconomic context; and sources of funding for garden activities.

Garden age/activity duration was: less than 1 year; 1-3 years; 3-7 years; or above 7 years.

Number of active members was: up to 10; 10-20; 20-30; 30-40; or above 40.

Annual visits were: up to 100; 100-500; 500-1,000; 1,000-5,000; 5,000-10,000; or above 10,000.

The annual-visit response referred to the estimated number of visits rather than the number of unique individuals and could therefore include repeated visits by the same person. Although the original form also requested a general assessment of neighbourhood socioeconomic level, we assigned the SEI variables used in the analyses independently from 2021 Israel Central Bureau of Statistics data.

Spatial-data technical specifications

We downloaded the spatial datasets used in the reported analyses on 12 May 2025. We obtained the national green-area layer, representing statutory land-use designations, and the road layer from Maps of Israel, accessed at the time through a publicly available Ministry of Education ArcGIS service. We obtained CBS statistical-area boundaries separately from the Israel Central Bureau of Statistics and linked them to the corresponding 2021 socioeconomic data using statistical-area identifiers (Israel Central Bureau of Statistics (CBS), 2024).

Garden locations were initially represented in WGS84 (EPSG:4326) and projected to Israel Transverse Mercator (EPSG:2039) before distance- and area-based calculations. We spatially intersected garden points with CBS statistical-area polygons to assign statistical-area identifiers and with national statutory green-area polygons to determine whether each garden was located within an area formally designated as green space. Two gardens did not intersect a residential statistical area-one was located in an industrial area and one within a park-and were assigned the corresponding city-average SEI.

We checked statutory green-area polygon geometries and repaired invalid geometries before area calculations. We calculated polygon areas in EPSG:2039 and summed them by municipality. We calculated city-level formal green-space extent by dividing the summed statutory green-area polygon area by the corresponding municipal area from Local Authorities in Israel, 2023 (Israel Central Bureau of Statistics (CBS), 2026).

For the street-network variable, we generated a 400-m Euclidean-radius buffer around each CG. We projected the road-line layer to EPSG:2039 and excluded the motorway, motorway_link, trunk, and trunk_link classes. We intersected the remaining road segments with each garden buffer, calculated their lengths in metres, and summed segment lengths for each CG to obtain total street-network length within 400 m.

Analytical variables

Supplementary Table S7 | Variables included in the dimension-specific models

Analytical dimension	Variables
Garden-level characteristics	Active-member ordinal variable; garden age/activity duration.
Funding sources	Municipal funding; community self-funding; government funding; local institutional funding; external funding.
Socioeconomic context	Neighbourhood SEI (1-10); city-average SEI.
Spatial/GIS context	Inside formally designated green area (statutory layer); Open green space (reported site type); recreation/leisure facility; care/welfare institution; walkable street length within 400 m; educational institution; religious institution; community institution; residential/neighbourhood setting.
Community profile	Educational community; residents; people with disabilities; youth; organised groups; religious groups; families; seniors; diverse communities.

Supplementary Table S8 | Proportional-odds assumption diagnostics for the dimension-specific ordinal regression models

Analytical dimension	Variable	Wald F	P value	Plm-adjusted P value	Evidence against proportional odds
Garden-level characteristics	Active members (ordinal)	1.82	0.204	0.279	No
Garden-level characteristics	Garden age/activity duration	2.33	0.139	0.279	No
Socioeconomic context	Neighbourhood SEI (1-10)	0.65	0.538	0.538	No
Socioeconomic context	City-average SEI	1.57	0.249	0.497	No
Funding sources	Municipal funding	49.37	<0.001	<0.001	Yes
Funding sources	Community self-funding	0.72	0.509	0.692	No
Funding sources	Government funding	7.46	0.008	0.031	Yes
Funding sources	Local institutional funding	1.81	0.205	0.616	No
Funding sources	External funding	1.16	0.346	0.692	No
Spatial/GIS context	Inside formally designated green area (statutory layer)	2.74	0.105	0.797	No
Spatial/GIS context	Street-network length within 400 m	0.81	0.468	1.000	No
Spatial/GIS context	Open green space (reported site type)	1.31	0.306	1.000	No

Spatial/GIS context	Community institution	0.10	0.907	1.000	No
Spatial/GIS context	Educational institution	0.69	0.519	1.000	No
Spatial/GIS context	Residential / neighbourhood setting	0.05	0.952	1.000	No
Spatial/GIS context	Care/welfare institution	4.63	0.032	0.291	No
Spatial/GIS context	Religious institution	2.81	0.100	0.797	No
Spatial/GIS context	Recreation/leisure facility	0.50	0.616	1.000	No
Community profile	Families	1.44	0.276	0.756	No
Community profile	Educational community	3.51	0.063	0.504	No
Community profile	Seniors	1.17	0.342	0.756	No
Community profile	Youth	2.30	0.142	0.756	No
Community profile	Residents	2.39	0.133	0.756	No
Community profile	Organised groups	2.51	0.123	0.756	No
Community profile	Diverse communities	2.31	0.142	0.756	No
Community profile	People with disabilities	2.70	0.108	0.756	No
Community profile	Religious groups	4.92	0.027	0.247	No

Note: We assessed the proportional-odds assumption using a Brant-style parallel-slopes diagnostic based on cumulative binary logistic regressions, with municipality-clustered robust inference consistent with the primary models. Holm-adjusted P values account for multiple variable-level tests within each analytical dimension. A significant adjusted P value indicates evidence that a variable's association differs across cumulative annual-visit thresholds; non-significant results indicate no evidence against the assumption, rather than proof that the assumption holds. We detected evidence against proportional odds for municipal funding and government funding only.

Supplementary Table S9 | Sensitivity analysis using a four-level annual-visit outcome

Analytical dimension	Variable	Primary six-level β	Primary P value	Four-level sensitivity β	Sensitivity P value	Same coefficient direction	Primary significant association retained
Garden-level characteristics	Active members (ordinal)	0.78	0.009	0.81	0.014	Yes	Yes
Garden-level characteristics	Garden age/activity duration	-0.02	0.900	-0.03	0.827	Yes	-
Socioeconomic context	Neighbourhood SEI (1-10)	0.14	0.623	0.14	0.625	Yes	-
Socioeconomic context	City-average SEI	0.30	0.191	0.31	0.197	Yes	-
Funding sources	Municipal funding	0.23	0.109	0.23	0.110	Yes	-

Funding sources	Community self-funding	0.57	<0.001	0.58	0.001	Yes	Yes
Funding sources	Government funding	0.27	0.022	0.27	0.022	Yes	Yes
Funding sources	Local institutional funding	0.03	0.847	0.01	0.951	Yes	-
Funding sources	External funding	0.21	0.026	0.20	0.030	Yes	Yes
Spatial/GIS context	Inside formally designated green area (statutory layer)	0.29	0.048	0.30	0.046	Yes	Yes
Spatial/GIS context	Street-network length within 400 m	0.05	0.682	0.06	0.617	Yes	-
Spatial/GIS context	Open green space (reported site type)	0.22	0.321	0.22	0.306	Yes	-
Spatial/GIS context	Community institution	-0.29	0.100	-0.29	0.108	Yes	-
Spatial/GIS context	Educational institution	0.04	0.787	0.04	0.817	Yes	-
Spatial/GIS context	Residential / neighbourhood setting	-0.30	0.018	-0.29	0.026	Yes	Yes
Spatial/GIS context	Care/welfare institution	0.12	0.126	0.13	0.118	Yes	-
Spatial/GIS context	Religious institution	-0.16	0.256	-0.15	0.325	Yes	-
Spatial/GIS context	Recreation/leisure facility	0.13	0.194	0.14	0.178	Yes	-
Community profile	Families	-0.02	0.884	-0.02	0.872	Yes	-
Community profile	Educational community	0.29	<0.001	0.26	0.003	Yes	Yes
Community profile	Seniors	-0.10	0.657	-0.13	0.579	Yes	-
Community profile	Youth	0.14	0.331	0.16	0.301	Yes	-
Community profile	Residents	0.23	0.042	0.22	0.036	Yes	Yes
Community profile	Organised groups	0.01	0.861	0.00	0.955	Yes	-
Community profile	Diverse communities	-0.23	0.107	-0.23	0.101	Yes	-
Community profile	People with disabilities	0.14	0.314	0.13	0.304	Yes	-
Community profile	Religious groups	0.01	0.902	0.02	0.850	Yes	-

Note: As a sensitivity analysis for sparse upper annual-visit levels, the three highest levels of the original six-level ordinal variable were combined into a single level representing $\geq 1,000$ annual visits. We analysed the resulting four-level outcome using the same dimension-specific proportional-odds models as in the primary analysis. All associations that were statistically significant in the primary six-level models retained the same direction and remained statistically significant in the four-level sensitivity analysis.